

**SECTION A**

General Unit Information:

Unit Name	Principles Of Programming And Data Structures - COPY
Unit Code	CIS096-1
Level	4
Credit Value	30
Location(s) of Delivery	On Campus and Off Campus
Period(s) of Delivery	11 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	Programming as well as the knowledge of commonly used data structures are fundamental skills for all computer science practitioners. In this unit we will transmit the programming skills required for a professional career in all computer science aspects. In particular, the aims of this unit are: 1. To develop programming skills using a high level programming language. 2. To explore the process of analysis, design, implementation and testing of systems. 3. To apply knowledge in respect of modelling to real world problems and implement them using a programming language. 4. To evaluate, critique and synthesise existing approaches to software development. 5. To present commonly used data structures
Syllabus Content	This unit builds students' understanding of programming fundamentals. Deeper issues in driving from modelling to programming environments and the interrelationship between the two are explored further. The core techniques associated with programming in an Object Oriented (OO) environment as well as commonly used data structures are explored in detail. Students will enhance their approach to problem solving, decomposition and abstraction strategies for finding solutions to problems. Good practice in analysis, design, implementation and testing will be built upon, adding to an understanding of how to produce high quality software.

Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Comprehend and use basic programming concepts and data structures related to high level programming within the object oriented programming paradigm.

2. Demonstrate the following skills and abilities

Apply knowledge and understanding of basic programming concepts and data structures to develop a software solution using a modern high level programming language

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	72	140	60	28	0	300
Percentage	24%	47%	20%	9%	0%	99%

Approach to Learning

The knowledge and ability in programming is essential to the study of computing. This unit develops programming skills in a combined and balanced theoretical and practical approach, which both will take up roughly half of the unit. The required theory will be taught in lectures and applied in practical sessions and assignments. The students need to exhibit a basic understanding of the taught fundamental concepts in a computer-based exam. Furthermore, to simulate typical situations in real-life software development and programming as well as for students to apply their knowledge in a challenging task, students will collaborate in a team and individually contribute to the design and develop of a major software project

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Summative in-class test or phase test	1 and 2	30	6	1 hour
2	Coursework - Artefact	1 and 2	70	13	na

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	You are actively engaged in enquiry during the learning process and you have to articulate your own understanding of the programming issues examined in the class.
Contextual Understanding, including where employer based	This unit allows you to understand different programming concepts and apply them in different domains
Collaboration	Students will develop their collaborative abilities through the development and presentation of a software system in a group.
Enterprise	The unit fosters understanding of commercial activity (designing a system matching a 'client' specification).

Assessment 1 Details	You will undertake a 1 hour in-class test that will cover core topics in computer programming, including data structures and their use
Assessment 2 Details	You will submit an artefact (piece of software) as specified in the format outlined in the detailed assessment brief for the unit that will use your knowledge and skills of computer programming and software development.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

Define basic programming concepts such as classes, variables and mathematical operators

Use conditional statements to make logical choices within a computer program

User while/for loops to allow a program to run a cycle of operations

In order to pass Assessment 2 you will need to:

Write a piece of programming code for a given scenario that solves the problem posed

Reflect upon your strength and weaknesses in relation to a programming project you have undertaken and use these to improve your program code

Outline a test routine to check the logical correctness of a program you have written and provide the results of your testing demonstrating a working piece of code

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

See <https://rl.talis.com/3/bedfordshire/lists/970D7182-B43D-B59A-FDC0-0E10D3B6D9F3.html>

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://rl.talis.com/3/bedfordshire/lists/970D7182-B43D-B59A-FDC0-0E10D3B6D9F3.html>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://rl.talis.com/3/bedfordshire/lists/970D7182-B43D-B59A-FDC0-0E10D3B6D9F3.html>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	Y	
The approach to practical work takes account of the needs of students with disabilities?	Y	
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	

Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	NA	
--	----	--

* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100956
Unit Co-ordinator	Susan Brandreth
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Cybersecurity
BSc (Hons) Computer Networking
BSc (Hons) Computer Science
BSc (Hons) Software Engineering
BSc (Hons) Business Information Systems
BSc (Hons) AI & Robotics
BSc (Hons) Information and Data Systems
BSc (Hons) Cybersecurity - Hong Bang International University (HBIU)
BSc (Hons) Cybersecurity - Middle East University (MEU)